

Smart Parking System — Circuit Design

Architecture

Four HC-SR04 sensors feed an ESP32. The ESP32 drives an SSD1306 OLED, eight slot LEDs, a buzzer, and a servo barrier demonstration.

Pin mapping

Function	GPIO / Connection
OLED SDA	21
OLED SCL	22
Servo PWM	18
Buzzer	19
S1 TRIG/ECHO	5 / 17
S2 TRIG/ECHO	16 / 4
S3 TRIG/ECHO	27 / 26
S4 TRIG/ECHO	25 / 33
S1 Green/Red	12 / 2
S2 Green/Red	14 / 13
S3 Green/Red	32 / 23
S4 Green/Red	15 / 0

LED behavior

Green = FREE. Red = OCCUPIED. Each LED uses a 220 Ω series resistor.

Sensor logic

Five readings are collected for each sensor. The highest and lowest readings are removed, and the remaining three are averaged. Average < 35 cm means OCCUPIED; otherwise FREE.

Physical safety

The virtual Wokwi diagram directly connects HC-SR04 ECHO to ESP32 GPIO for simulation. For physical construction, use an appropriate voltage divider/level shifter because the ESP32 GPIO is not a 5 V-tolerant input. Power the servo appropriately and maintain common ground.

Virtual-only pin note

GPIO0 is used for Slot 4 red LED in the supplied virtual circuit. It should be replaced with a safer output GPIO in a physical implementation.